Signals and Systems
SET214
Lecture (07)- Continuous-Linear Time Invariant (LTI) systems

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Outlines

- Linear Time-Invariant
- Convolution
- Exercise
- Convolution Properties
- Block Diagram Representation
- Exercise

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Linear Time-Invariant

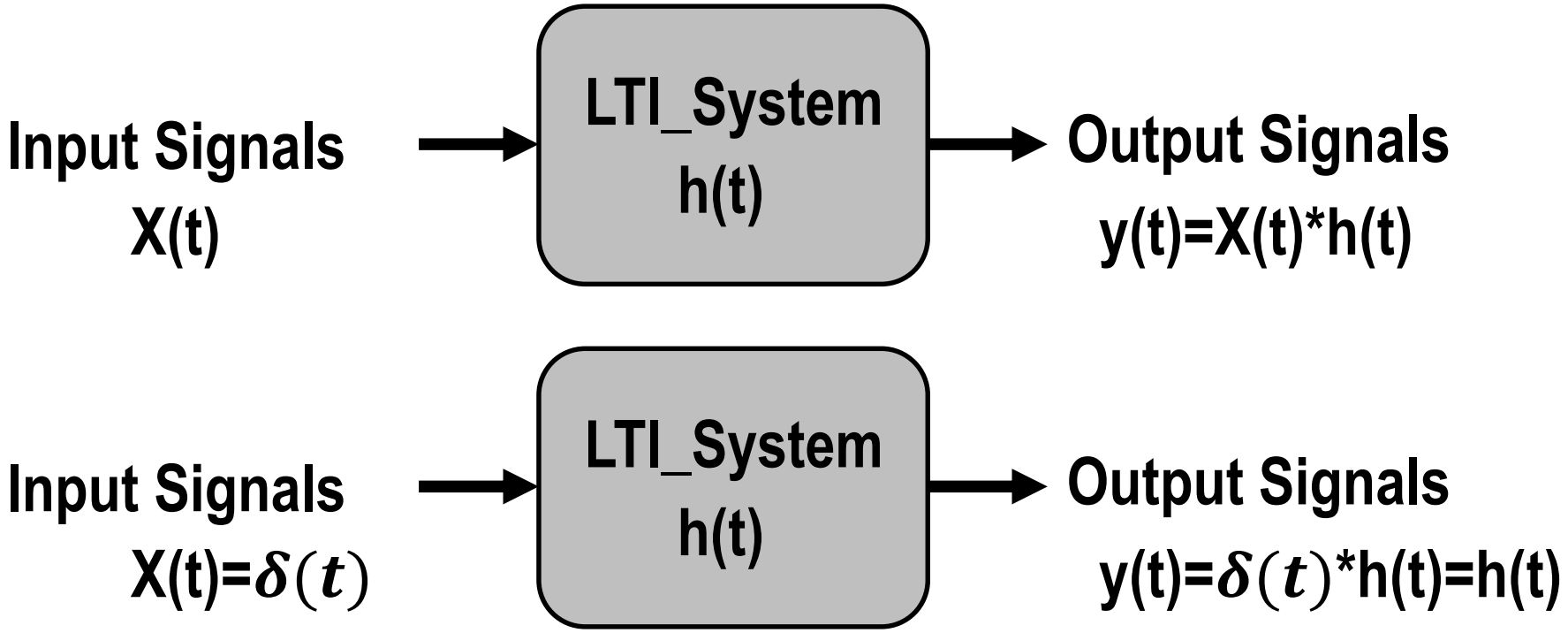
- Linear Time-Invariant (LTI) system is a system that satisfies both linearity and time-invariance Properties (sometimes referred to as a Linear Shift-Invariant (LSI) system.
- In engineering, linear time-invariant (LTI) systems play a very important role.
- Very powerful mathematical tools have been developed for analyzing LTI systems.→ convolution.
- In practice, systems that are not LTI can be well approximated using LTI models.
- LTI systems are much easier to analyze than systems that are not LTI

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Convolution (1/3)

- **Convolution** is the process by which an **input** interacts with an LTI system to produce an **output**.
- Consider an LTI system with **impulse response** →h(t) whose input and output are x(t) and y(t), respectively



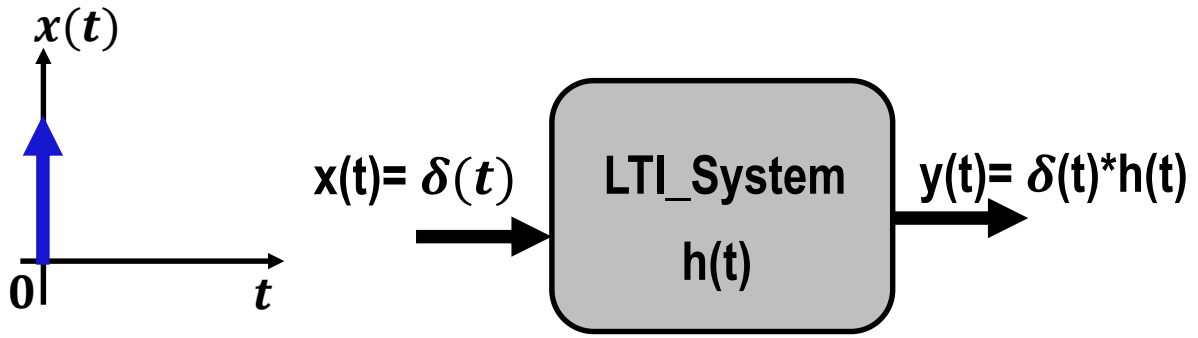
If input to LTI system is **unit impulse** then output of system is known as **impulse response**

Convolution (2/3)

Note:-

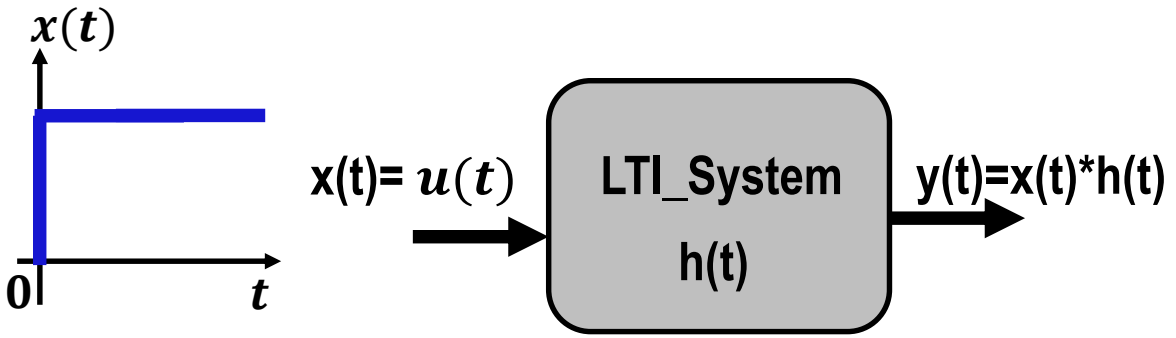
- **When the input to LTI system is impulse then the output of LTI system is known as impulse response**

$$y(t)= \delta(t)*h(t)=h(t)$$



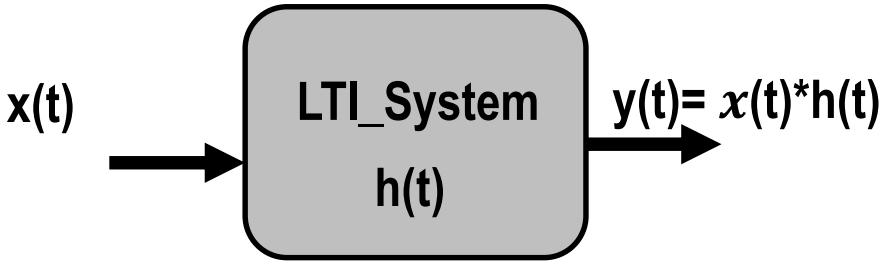
- Similarly for the input to LTI system is **unit step** signal then the output of LTI system is known as **step response**.

$$y(t)=u(t)*h(t)$$



Convolution (3/3)

- Convolution is a mathematical operator & it is used for calculation of response of LTI system.
- Herein, the asterisk symbol (i.e., "$*$") will always be used to denote convolution, not multiplication.



$y(t) = x(t) * h(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

Steps for Graphical Convolution

1- Re-Write the signals as functions of τ : $x(\tau)$, $h(\tau)$

2- Flip $h(\tau)$ around $t = 0$ to get $h(-\tau)$. Usually easier to flip the shorter signal

3-Shift $h(-\tau)$ by an arbitrary value of t to get $h(t-\tau)$ and get its edges.

4-Find regions of overlap. For each region: form the product $x(\tau)\ h(t-\tau)$ and Integrate.

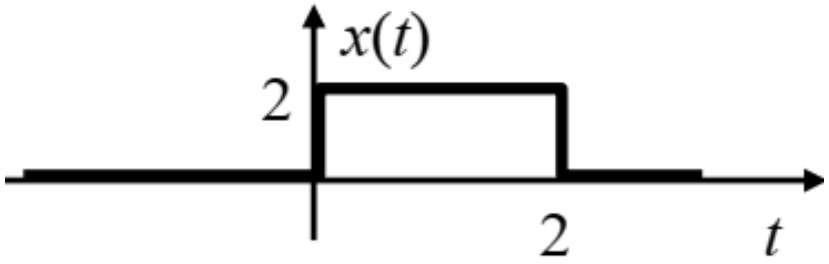
5- Plot the result of convolution

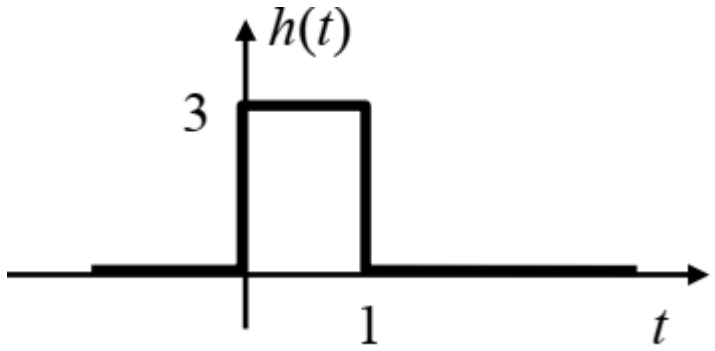
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Exercise (1/5)

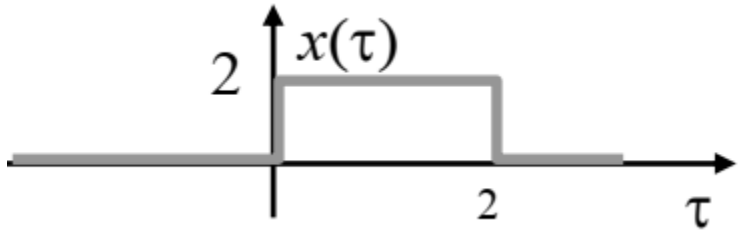
- Convolve these two signals: y(t)=x(t)*h(t)

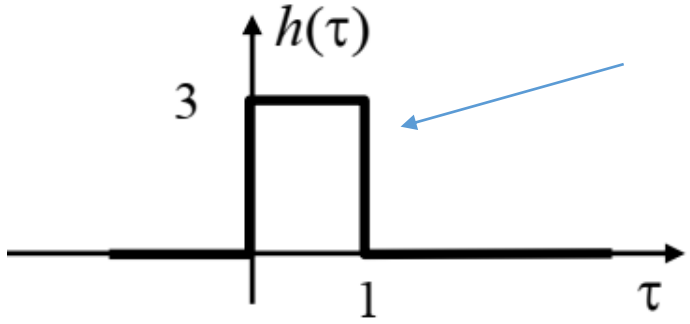




Ans.

1- Re-Write the signals as functions of τ : x(τ), h(τ)

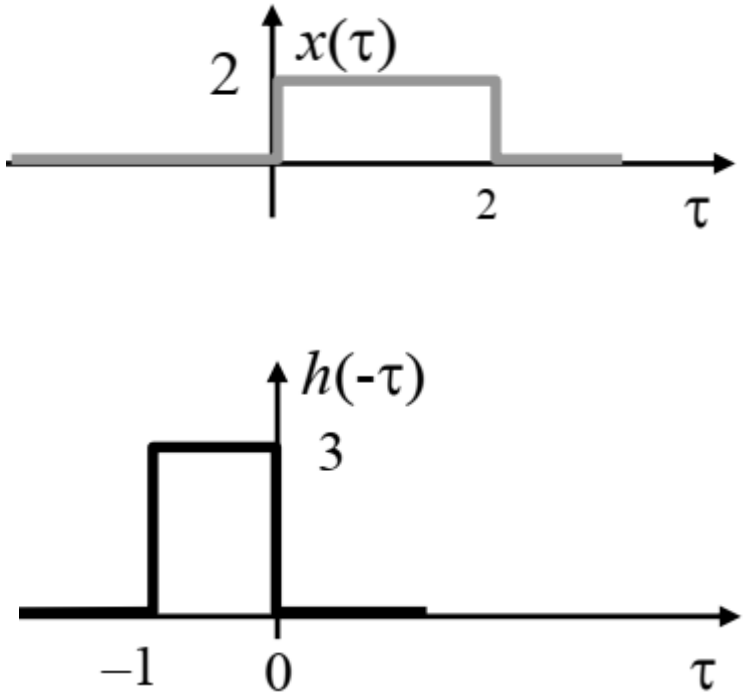




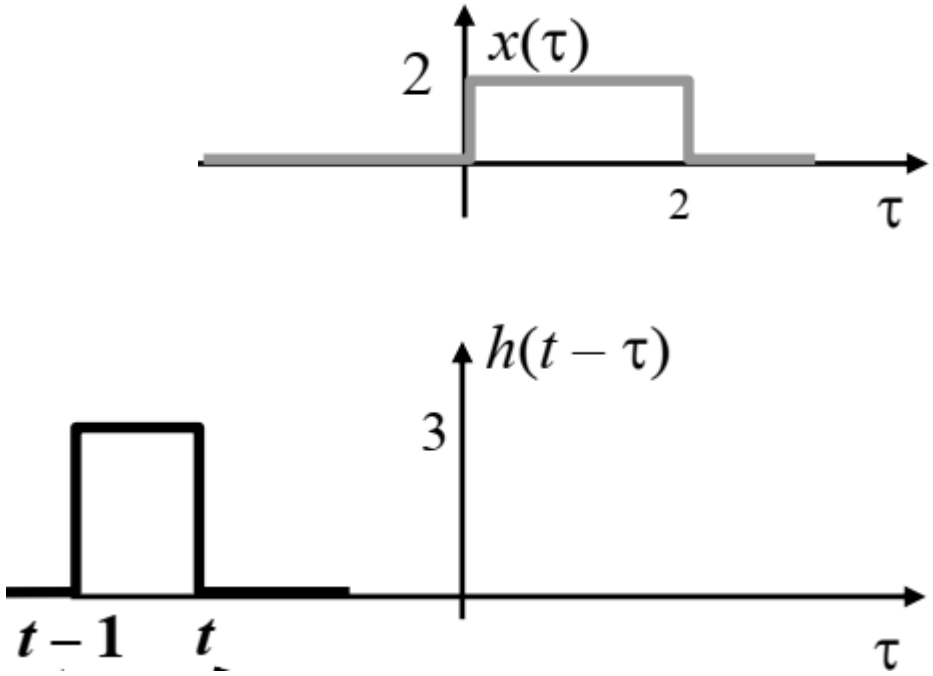
Shorter signal

Exercise (2/5)

2- Flip h(τ) around t = 0 to get $h(-\tau)$. Usually easier to flip the shorter signal



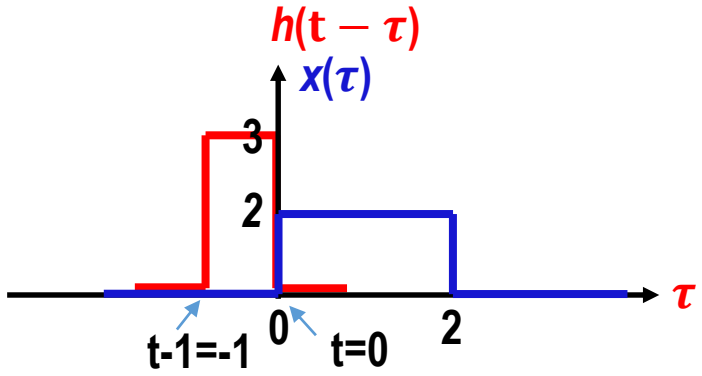
3-Shift $h(-\tau)$ by an arbitrary value of t to get $h(t-\tau)$ and get its edges.



Exercise (3/5)

4-Find regions of overlap. For Each Region: form the product $x(\tau)\ h(t-\tau)$ and Integrate

❑ Region (1) → t =0

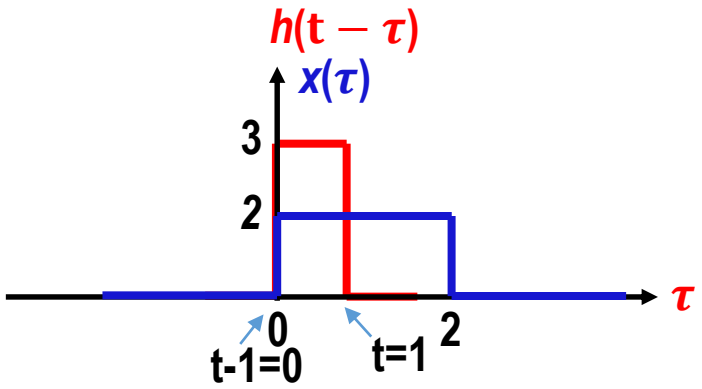


$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

There is not overlap between $h(t-\tau)$ and $x(\tau)$

y(t)=0 for all t< 0

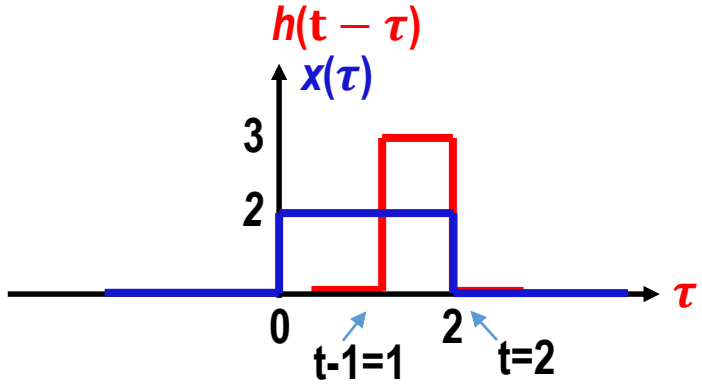
❑ Region (2) → t =1



$$y(t) = \int_{0}^{1} 2 \,.\, 3\, d\tau = 6$$

y(t)=6 @ t= 1

❑ Region (3) → t = 2



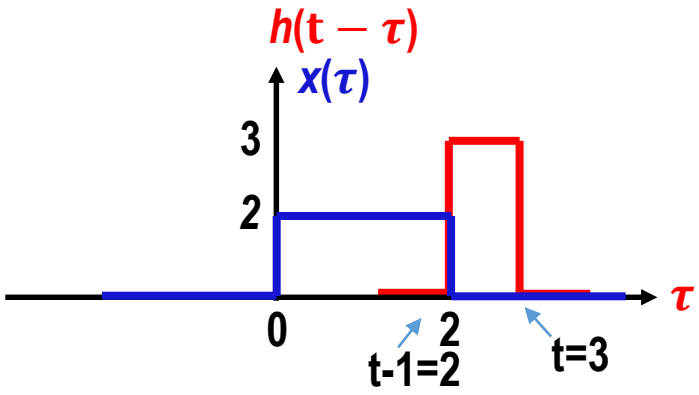
$$y(t) = \int_{1}^{2} 2 \,.\, 3\, d\tau = 6$$

y(t)=6 @ t =2

Exercise (4/5)

4-Find Regions of Overlap. For Each Region: Form the Product $x(\tau)$ $h(t-\tau)$ and Integrate

❑ Region (4) → t =3

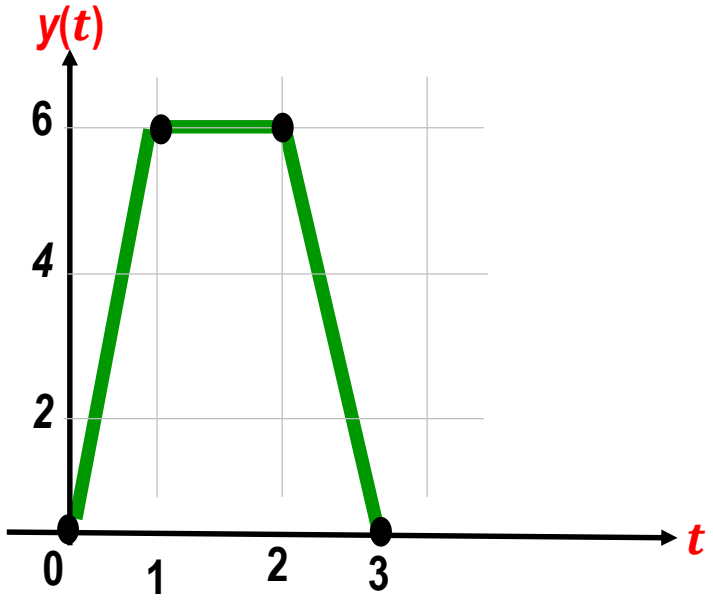


There is not overlap between h(t − τ) and x(τ)

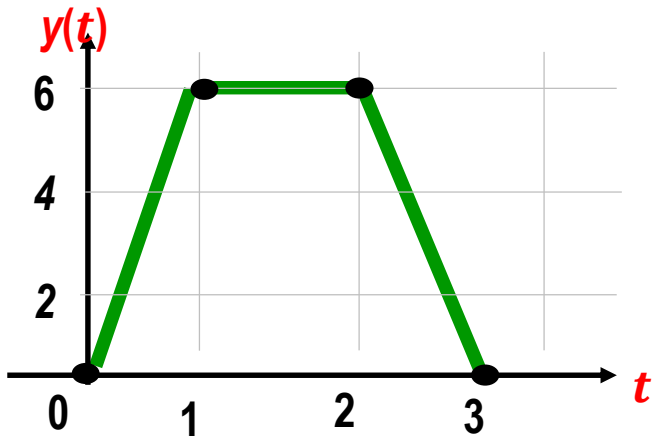
y(t)=0 @ t=3

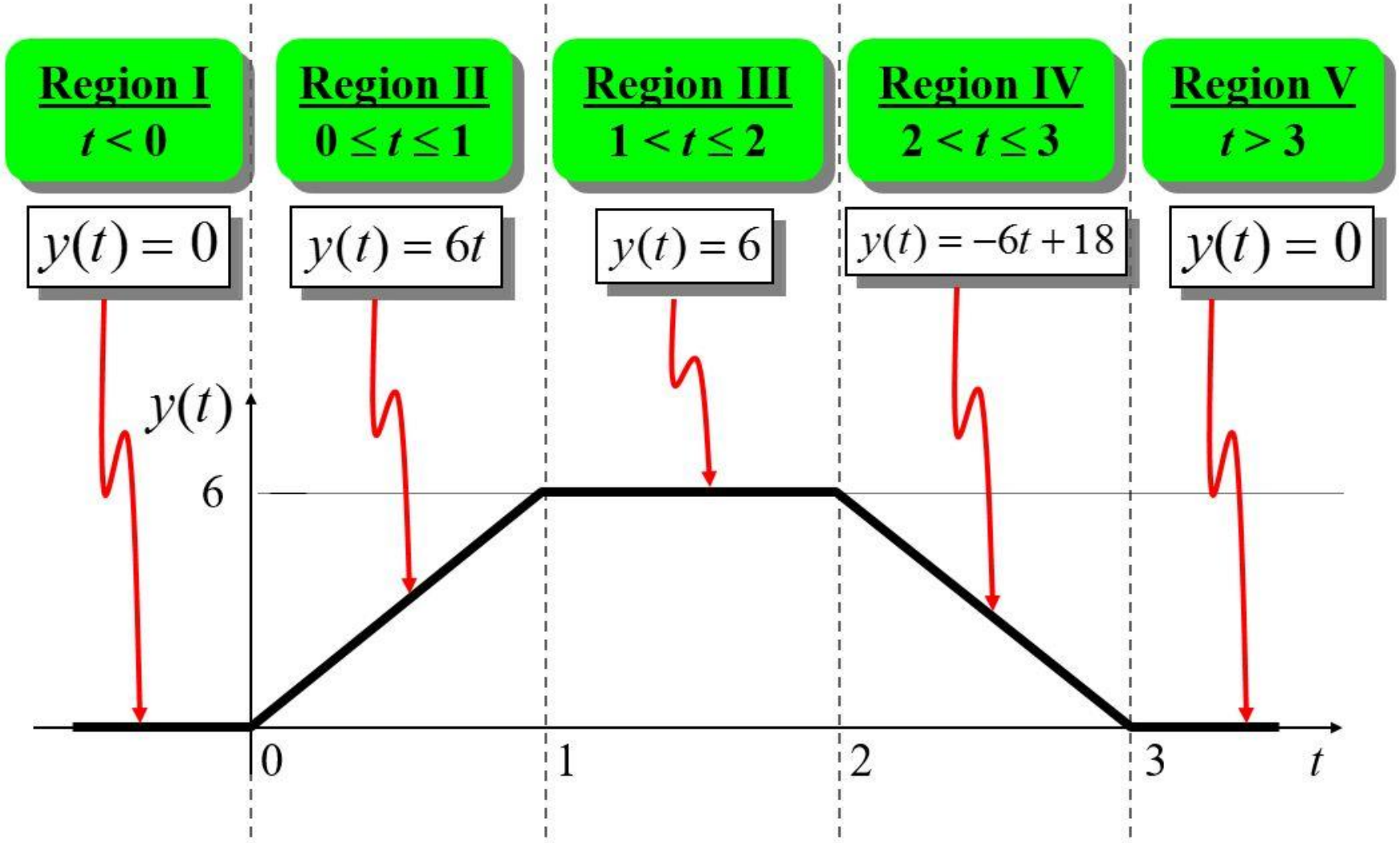
5- Plot the result of convolution

t	y(t)
0	0
1	6
2	6
3	0



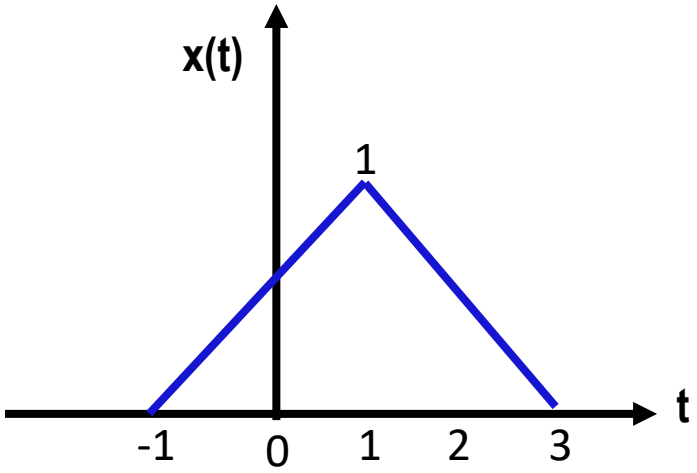
Exercise (5/5)

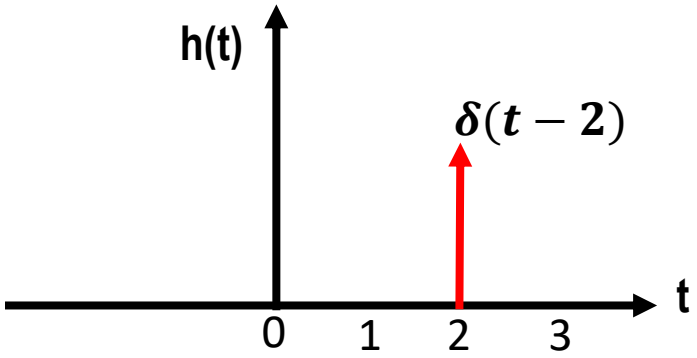

y(t)
6
4
2
0
1
2
3
t

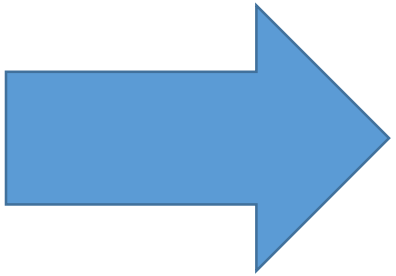

Region I
t < 0
Region II
0 ≤ t ≤ 1
Region III
1 < t ≤ 2
Region IV
2 < t ≤ 3
Region V
t > 3
y(t) = 0
y(t) = 6t
y(t) = 6
y(t) = −6t + 18
y(t) = 0
y(t)
6
0
1
2
3
t

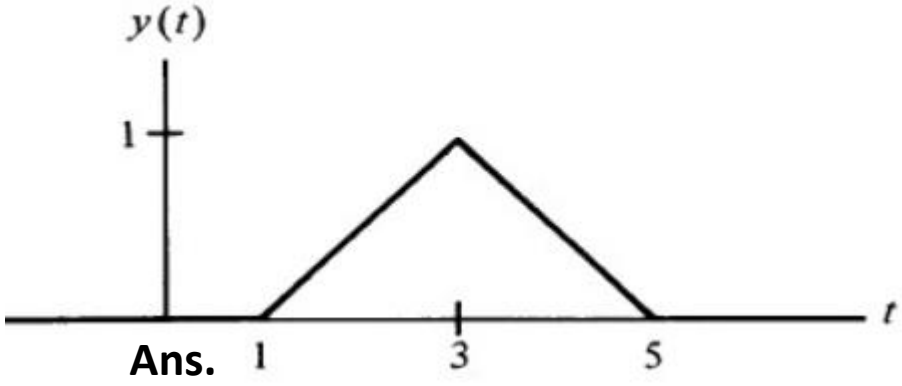
Exercise

- Convolve these two signals: y(t)=x(t)*h(t)









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Convolution Properties

- Since convolution is frequently employed in the study of LTI systems, it is important for us to know some of its basic properties.

Commutative

$$x(t)*h(t) = h(t)*x(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

Associative

$$x_1(t)*[h_1(t)*h_2(t)] = [x_1(t)*h_1(t)]*h_2(t)$$

Distributive

$$X(t)*(h_1(t) + h_2(t)) = X(t)*h_1(t) + X(t)*h_2(t)$$

Impulse Response

$$x(t)*\delta(t) = x(t) \Rightarrow x(t)*\delta(t - t1) = x(t - t1)$$

Ex.,

$$u(t-1)*\delta(t+2) = u[(t+2)-1] = u(t+1)$$

Step Response

- The response of a system to the input *u(t)* is called the **step response** of the system

$$h(t) = \frac{ds(t)}{dt}$$

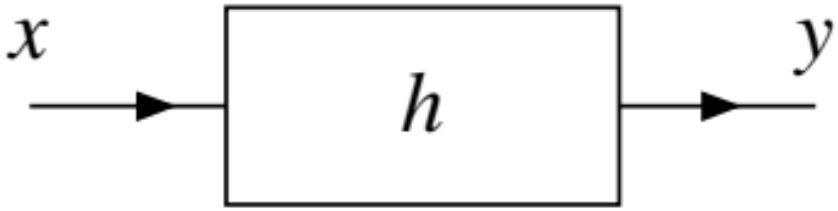
The step response provides a practical means for determining the impulse response of a system

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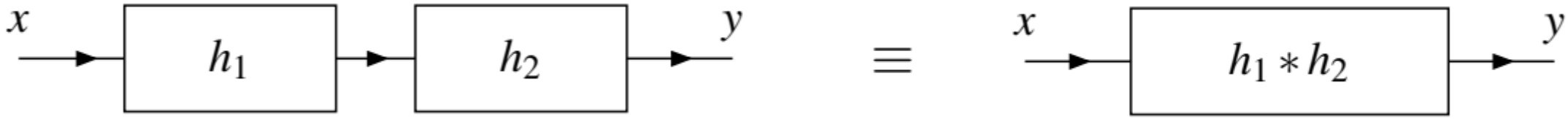
Block Diagram Representation (1/2)

- Often, it is convenient to represent a (CT) LTI system in block diagram form.
- Since such systems are completely characterized by their impulse response, we often label a system with its impulse response
- That is, we represent a system with input *x*, output *y*, and impulse response *h*, as shown below.



Block Diagram Representation (2/2)

- The series interconnection of the LTI systems with impulse responses h1 and h2 is the LTI system with impulse response h1∗h2. That is, we have the equivalence shown below.



- The *parallel* interconnection of the LTI systems with impulse responses *h*1 and *h*2 is the LTI system with impulse response *h*1 +*h*2. That is, we have the equivalence showr

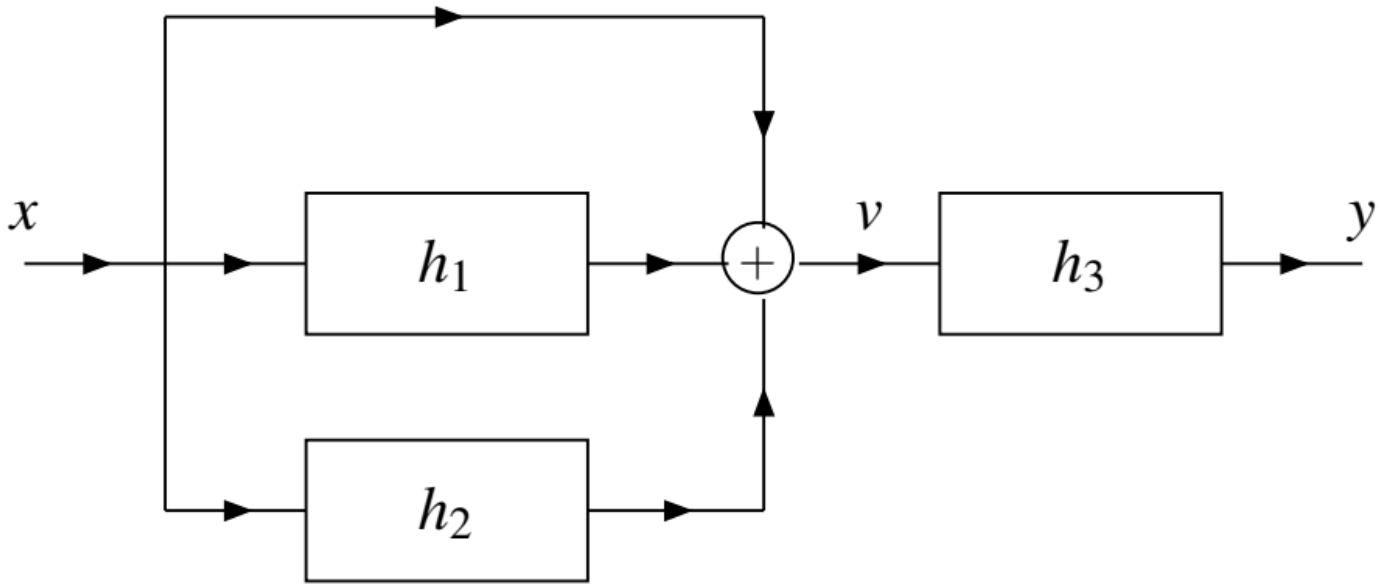


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Exercise (1/2)

- Consider the system with input x, output y, and impulse response h as shown. Each subsystem in the block diagram is LTI and labelled with its impulse response. Find h.



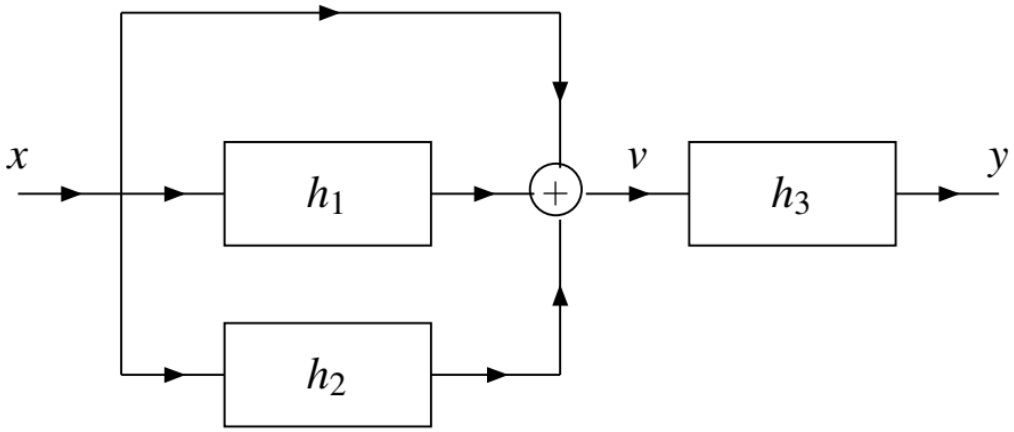
Exercise (2/2)

- From the left half of the block diagram, we can write

$v(t) = x(t)+x(t)*h1(t)+x(t)*h2(t)$

$= x(t)*\delta(t)+x(t)*h1(t)+x(t)*h2(t)$

Where, $X(t)*\delta(t)= X(t)$



$v= x * [\delta+h1+h2]$

$y = v*h3$

$$y = x * [\delta+h1+h2]*h3$$

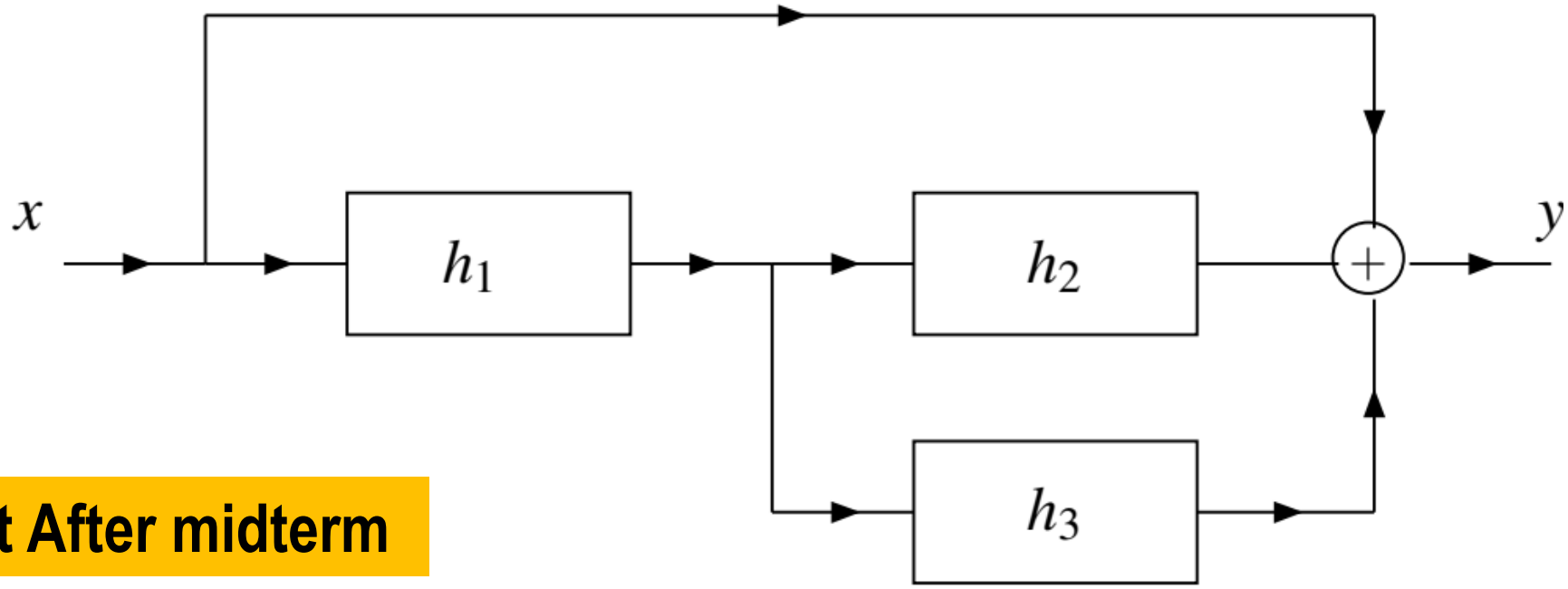
$$y = x * [\delta*h3+h1*h3+h2*h3]$$

$$y = x * [h3+h1*h3+h2*h3]$$

$$h = [h3+h1*h3+h2*h3]$$

Assignment (1)

- Consider the system with input *x* and output *y* as shown in the figure below. Each system in the block diagram is LTI and labelled with its impulse response. Find the impulse response *h* of the overall system in terms of *h*1, *h*2, and *h*3



Assignment After midterm

Thanks!